Math708 – Foundations of Computational Mathematics I

Fall 2012

Homework 4

1. Consider the Trapezoidal scheme

$$u_{n+1} = u_n + \frac{h}{2}(f_n + f_{n+1})$$

for the solution of the initial value problem

$$y' = f(t, y), \quad y(0) = y_0, \quad t \in [0, T]$$

Assume that the right hand side f is Lipschitz continuous with respect to y. Show that for sufficiently small step size h, the Trapezoidal scheme has a unique solution u_{n+1} for any u_n.

2. Consider the multi-step scheme

$$u_{n+1} - u_{n-1} = h(f_{n+1} - 3f_n + 4f_{n-1})$$

for solving the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0.$$

Compute its order. Is it stable?

3. Consider solving the initial value problem

$$y' = f(t, y), \quad y(0) = y_0, \quad t \in [0, 1]$$

by the following method

$$u_{n+1} = \alpha u_n + \beta u_{n-1} + h\gamma f_{n-1}, \quad n \geq 1,$$

$$u_0 = y_0, \quad y_1 = y_0 + hf(t_0, y_0).$$

Choose the constants α, β, and γ so that the order of the method is as high as possible. Determine whether the resulting method is convergent.

4. Consider the initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

Prove that an explicit, 3-stage, third order Runge-Kutta method of the form

$$u_{n+1} = u_n + h(b_1K_1 + b_2K_2 + b_3K_3)$$

with

$$\begin{aligned} K_1 &= f(t_n, u_n) \\ K_2 &= f(t_n + c_2 h, u_n + h a_{21} K_1) \\ K_3 &= f(t_n + c_3 h, u_n + h a_{31} K_1 + h a_{32} K_2) \end{aligned}$$

must satisfy the following equations:

$$\begin{aligned} & b_1 + b_2 + b_3 = 1 \\ & c_2 = a_{21} \\ & c_3 = a_{31} + a_{32} \\ & b_2 c_2 + b_3 c_3 = \frac{1}{2} \\ & b_2 c_2^2 + b_3 c_3^2 = \frac{1}{3} \\ & b_3 a_{32} c_2 = \frac{1}{6} \end{aligned}$$

5. Consider the system of ODEs

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t), \qquad \mathbf{y}(0) = (1, 0, -1)^T$$

where $\mathbf{y}(t) = (y_1(t), y_2(t), y_3(t))^T$ and

$$\mathbf{A} = \begin{pmatrix} -21 & 19 & -20 \\ 19 & -21 & 20 \\ 40 & -40 & -40 \end{pmatrix}.$$

The exact solution is given by

$$\begin{cases} y_1(t) = \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-40t}(\cos 40t + \sin 40t) \\ y_2(t) = \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-40t}(\cos 40t + \sin 40t) \\ y_3(t) = -e^{-40t}(\cos 40t - \sin 40t) \end{cases}$$

Numerically solve this ODE system for $t \in [0, 2]$ using the standard 4-th order RK method with the stepsizes i) $h = 0.08$, ii) $h = 0.04$, and iii) $h = 0.02$. Compare your simulations with the exact solution and plot the errors for each component y_i.